

Band 7–8 Digital Technologies 2026

Curriculum and assessment plan

St Brendan's College (Yeppoon)



Level description	Context and cohort considerations
By the end of Year 8 students should have had the opportunity to apply computational thinking by defining and decomposing real-world problems, creating user experiences, designing and modifying algorithms, and implementing them in a general-purpose programming language. This involves students practising problem decomposition, using approaches such as divide and conquer to more clearly understand a problem by describing its component parts. Students represent and communicate their algorithmic solutions using flowcharts and pseudocode. Students check their solutions meet the specifications by testing and debugging their algorithms before and during implementation. They develop a deeper understanding of abstraction by explaining how and why digital systems represent data as whole numbers, which are then represented in binary. Students build on their skills from Mathematics (<i>Statistics</i>) in acquiring and interpreting data. In Digital Technologies, students continue to advance these skills and are also given opportunities to validate the data they acquire to ensure it is accurate and consistent. They collect and transform many types of data from a wide range of sources. Students model structured data in meaningful ways using spreadsheets and single-table databases, and analyse and visualise the data to extract meaning from it. They apply design thinking by using divergent techniques, such as mind mapping, role-play and using graphic organisers, to generate design ideas for user experiences and solution designs. Students review these ideas against design criteria and created user stories throughout their implementation as general-purpose programming by assessing them against current and future needs. They extend the use of these design criteria and user stories to evaluate the future impact of existing solutions. Students apply systems thinking by exploring the connections between hardware capabilities and tasks users want to perform. They investigate how data is transmitted via wired and wireless networks and explain the need for encryption to protect and secure data. Students use an increasing range of the features of digital tools to improve their efficiency and the consistency of the content they create, locate and communicate. They plan and manage projects individually and collaboratively, improving their control over the quality of their content. Students investigate personal security controls, including multi-factor authentication, to protect their data if passwords are compromised, and they understand the impact of phishing and other cyber security threats on people and data. In Digital Technologies, students should have frequent opportunities for authentic learning by making key connections with other learning areas.	

	Unit 1 — Digital Literacy	Unit 2 — Wearable Technology	Unit 3 — Cardbaord Arcade
Year level	7	8	8
Duration	1 Term	1 Semester	1 Semester
Description	In this unit, students develop foundational digital literacy skills by learning to confidently and responsibly use common digital tools and systems. Through practical, hands-on activities, students explore Microsoft Word, PowerPoint and Excel to create, manage and communicate information using appropriate conventions. The unit also introduces key concepts in cybersecurity, digital footprints and emerging technologies such as artificial intelligence, helping students understand how data is created, shared and protected in networked environments. Across the term, students build skills in selecting suitable digital tools, managing files and data, and using technology safely and ethically, preparing them for further learning in Digital Technologies and other learning areas.	In this unit, students use Micro:bit devices to explore data, communication and physical computing in a practical and age-appropriate context. Students learn how Microbits collect, transmit and represent data, with a focus on radio communication, binary representation and basic data security concepts. Through guided investigations, students explore how data can be gathered from sensors, transmitted between devices and analysed to identify patterns and trends. In the second part of the unit, students apply these skills to design and create wearable technologies, combining computational thinking, data analysis and ethical considerations when collecting data about people.	In this unit, students explore physical computing by learning to program Arduino microcontrollers and apply these skills to the design and construction of interactive arcade machines. Students develop an understanding of algorithms, basic programming structures, and how hardware specifications influence performance and functionality. The design process is emphasised through the use of user stories and design criteria, guiding students to create solutions that meet specific user needs. Through iterative design, testing and refinement, students combine computational thinking with creativity, problem-solving and hands-on construction skills.
	Assessment 1 — Digital Tasks	Assessment 2 — Data Exam	Assessment 4 — Hardware and Algorithms Exam
Term	1	1	3
Week	Week 10	Week 7	Week 6
Description	Students will create a digital portfolio that demonstrates their learning across the term through a collection of completed tasks and reflections. The portfolio includes evidence of using word processing, presentation and spreadsheet tools to create, organise, analyse and communicate information, as well as activities focused on cybersecurity awareness, safe online behaviour and responsible	In this assessment, students complete a written exam that assesses their understanding of how Microbits communicate and represent data. Students explain the differences between broadcast and direct radio communication, and demonstrate their understanding of how text, numbers and messages are represented using integers and binary. The exam also assesses students' ability to model and	In this assessment, students complete a written exam that assesses their understanding of foundational algorithms and hardware selection in the context of Arduino systems. Students demonstrate their ability to define and decompose problems, design and trace simple algorithms, and predict program outputs. The exam also evaluates students' understanding of how hardware specifications affect

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	use of AI tools. Students are expected to select and use digital tools efficiently, apply common conventions, manage data accurately, and demonstrate an understanding of how information is shared and secured online. The portfolio provides ongoing evidence of skill development, responsible digital practices and the ability to reflect on learning progress over time, aligned to the Year 7–8 Digital Technologies standards.	reason about structured data, design and trace basic algorithms, and predict program outputs related to Microbit communication and simple encryption. This assessment provides evidence of students’ conceptual understanding of data transmission, representation and algorithmic thinking.	performance and how to select appropriate components for specific tasks. This assessment provides evidence of students’ theoretical knowledge and computational thinking skills that underpin practical digital solutions.
Technique	Other Portfolio	Examination	Examination
Mode	Multimodal	Written	Written
Conditions	10 A4 PowerPoint slides that may include annotated graphical representations or equivalent media pages	- up to 50 minutes plus 10 minutes planning time - under supervised conditions - short response up to 75 words per item, up to 400 words for the task - extended response 100–200 words per item, up to 400 words for the task	- up to 50 minutes plus 10 minutes planning time - under supervised conditions - short response up to 75 words per item, up to 400 words for the task - extended response 100–200 words per item, up to 400 words for the task 1–2 A3 pages or equivalent digital media that may include graphical representations with annotations
Achievement standard	By the end of Year 8 students develop and modify creative digital solutions, decompose real-world problems, and evaluate alternative solutions against user stories and design criteria. Students acquire, interpret and model data with spreadsheets and represent data with integers and binary. They design and trace algorithms and implement them in a general-purpose programming language. Students select appropriate hardware for particular tasks, explain how data is transmitted and secured in networks, and identify cyber security threats. They select and use a range of digital tools efficiently and responsibly to create, locate and share content; and to plan, collaborate on and manage projects. Students manage their digital footprint.	By the end of Year 8 students develop and modify creative digital solutions, decompose real-world problems, and evaluate alternative solutions against user stories and design criteria. Students acquire, interpret and model data with spreadsheets and represent data with integers and binary. They design and trace algorithms and implement them in a general-purpose programming language. Students select appropriate hardware for particular tasks, explain how data is transmitted and secured in networks, and identify cyber security threats. They select and use a range of digital tools efficiently and responsibly to create, locate and share content; and to plan, collaborate on and manage projects. Students manage their digital footprint.	By the end of Year 8 students develop and modify creative digital solutions, decompose real-world problems, and evaluate alternative solutions against user stories and design criteria. Students acquire, interpret and model data with spreadsheets and represent data with integers and binary. They design and trace algorithms and implement them in a general-purpose programming language. Students select appropriate hardware for particular tasks, explain how data is transmitted and secured in networks, and identify cyber security threats. They select and use a range of digital tools efficiently and responsibly to create, locate and share content; and to plan, collaborate on and manage projects. Students manage their digital footprint.
Moderation	Monitoring: Week 7 Conferencing: Week 9	Cross marked within the department	Cross marking to be completed within the department
		Assessment 3 — Wearable Tech	Assessment 5 — Arcade Machine
Term		2	4
Week		Week 9	Week 9
Description		In this assessment, students design and develop a wearable Microbit-based solution to collect and analyse data related to a given investigation task. Students plan how data will be gathered using appropriate sensors, design algorithms to collect and store the data, and use digital tools to analyse and visualise results to draw conclusions. They select suitable hardware, design the user experience, manage the project process and evaluate their solution against design criteria and user needs. Students also demonstrate responsible and ethical data collection practices, reflecting on the purpose and impact of their digital solution.	In this assessment, students design and build a functioning cardboard arcade machine based on a provided user profile and set of user stories. Students apply the design process to define requirements, establish design criteria, and plan a solution that meets the needs of the target user. They design the user experience, develop and test algorithms, and implement programs using Arduino hardware and sensors. Students iteratively refine their designs, evaluate their solutions against the design criteria and user stories, and communicate their ideas using appropriate digital tools. The project demonstrates students’ ability to integrate hardware selection, programming, creative design and project management to develop a complete digital solution.
Technique		Investigation	Project
Mode		Multimodal	Multimodal

	Unit 1 — Digital Literacy	Unit 2 — Wearable Technology	Unit 3 — Cardbaord Arcade
Conditions		2–4 A3 pages or equivalent digital media pages that may include annotated graphical representations	2–4 A3 pages or equivalent digital media pages that may include annotated graphical representations
Achievement standard		By the end of Year 8 students develop and modify creative digital solutions, decompose real-world problems, and evaluate alternative solutions against user stories and design criteria. Students acquire, interpret and model data with spreadsheets and represent data with integers and binary. They design and trace algorithms and implement them in a general-purpose programming language. Students select appropriate hardware for particular tasks, explain how data is transmitted and secured in networks, and identify cyber security threats. They select and use a range of digital tools efficiently and responsibly to create, locate and share content; and to plan, collaborate on and manage projects. Students manage their digital footprint.	By the end of Year 8 students develop and modify creative digital solutions, decompose real-world problems, and evaluate alternative solutions against user stories and design criteria. Students acquire, interpret and model data with spreadsheets and represent data with integers and binary. They design and trace algorithms and implement them in a general-purpose programming language. Students select appropriate hardware for particular tasks, explain how data is transmitted and secured in networks, and identify cyber security threats. They select and use a range of digital tools efficiently and responsibly to create, locate and share content; and to plan, collaborate on and manage projects. Students manage their digital footprint.
Moderation		Monitoring: Week 6 Conferencing: Week 8	Monitoring: Week 6 Conferencing: Week 8

Content descriptions	Units			Content descriptions	Units		
Knowledge and understanding	1	2	3	Processes and production skills	1	2	3
Digital systems explain how hardware specifications affect performance and select appropriate hardware for particular tasks and workloads (AC9TDI8K01)		✓	✓	Acquiring, managing and analysing data acquire, store and validate data from a range of sources using software, including spreadsheets and databases (AC9TDI8P01)	✓	✓	
investigate how data is transmitted and secured in wired and wireless networks including the internet (AC9TDI8K02)	✓	✓	✓	analyse and visualise data using a range of software, including spreadsheets and databases, to draw conclusions and make predictions by identifying trends (AC9TDI8P02)	✓	✓	
Data representation investigate how digital systems represent text, image and audio data using integers (AC9TDI8K03)	✓	✓		model and query the attributes of objects and events using structured data (AC9TDI8P03)		✓	
explain how and why digital systems represent integers in binary (AC9TDI8K04)		✓		Investigating and defining define and decompose real-world problems with design criteria and by creating user stories (AC9TDI8P04)		✓	✓
				Generating and designing design algorithms involving nested control structures and represent them using flowcharts and pseudocode (AC9TDI8P05)		✓	✓
				trace algorithms to predict output for a given input and to identify errors (AC9TDI8P06)		✓	✓
				design the user experience of a digital system (AC9TDI8P07)		✓	✓
				generate, modify, communicate and evaluate alternative designs (AC9TDI8P08)	✓	✓	✓
				Producing and implementing implement, modify and debug programs involving control structures and functions in a general-purpose programming language (AC9TDI8P09)		✓	✓
				Evaluating evaluate existing and student solutions against the design criteria, user stories and possible future impact (AC9TDI8P10)		✓	✓

Content descriptions	Units			Content descriptions	Units		
Knowledge and understanding	1	2	3	Processes and production skills	1	2	3
				Collaborating and managing select and use a range of digital tools efficiently, including unfamiliar features, to create, locate and communicate content, consistently applying common conventions (AC9TDI8P11)	✓	✓	✓
				select and use a range of digital tools efficiently and responsibly to share content online, and plan and manage individual and collaborative agile projects (AC9TDI8P12)	✓	✓	✓
				Privacy and security explain how multi-factor authentication protects an account when the password is compromised and identify phishing and other cyber security threats (AC9TDI8P13)	✓		
				investigate and manage the digital footprint existing systems and student solutions collect and assess if the data is essential to their purpose (AC9TDI8P14)		✓	

General capabilities	Units		
	1	2	3
Critical and creative thinking	✓	✓	✓
Digital literacy	✓	✓	✓
Ethical understanding		✓	
Intercultural understanding			
Literacy	✓	✓	
Numeracy	✓	✓	✓
Personal and social capability			

Cross-curriculum priorities	Units		
	1	2	3
Aboriginal and Torres Strait Islander histories and cultures			
Asia and Australia's engagement with Asia			
Sustainability			

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